Mark schemes

Q1.

(a) Putting $\frac{t^3}{216} = 0.9$

M1

$t = 5.793$

5.79 to 5.80

A1

41 days.

Accept 40 days in this context

A1

3

(b) Attempt to differentiate F(t)

ct^2 seen

M1

$f(t) = \frac{1}{72}t^2 \quad 0 \le t \le 6$

Condone domain missing here

A1

$= 0 \quad$ otherwise

*For **complete** function*

A1

3

(c) Attempt to integrate $t\text{f}(t)$ from 0 to 6

*Using their f(t) from **(b)** ct^4 seen*

M1

$E(T) = 4.5$

A1

Attempt to integrate $t^2\text{f}(t)$ from 0 to 6

*Using their f(t) from **(b)** ct^5 seen*

M1

$E(T^2) = 21.6$

A1

$\text{Var}(T) = \text{E}(T^2) - \text{E}(T)^2$

Applied in this case. Dependent on both M1

m1

$= 21.6 - 4.5^2 = 1.35$

A1

6

(d) S.d. $= \sqrt{1.35} = 1.162$

For $\sqrt{}$ (their Var) $0 < \text{Var}(T) < 9$

M1

Use of F(5.662)

For F(their s.d. + their E(T)) provided 0 < Total < 6

m1

$$1 - \frac{5.662^3}{216}$$

m1

$= 0.160$

AWFW 0.159 to 0.161

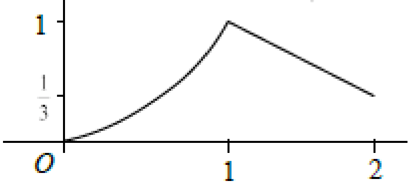
A1

4

[16]

Q2.

(a)



Curve concave upwards between (0, 0) and (1, y_1)

B1

Negative gradient line between (1, y_1) and (2, y_2) with y_2 > 0 (and not beyond 2)

B1

y_1 = 1 and y_2 = ⅓ shown

B1

3

(b) (i) Attempt to integrate t^2 **between 0 and x**

Accept integral of x^2

M1

$F(x) = \frac{1}{3}x^3$

A1

2

(ii) Their $F(x) = 0.25$

M1

$x = 0.909$

AWRT; accept $\sqrt[3]{0.75}$ OE

A1

2

(c) (i) $F(1) = \frac{1}{3}$

B1

$\int_1^x \frac{1}{3}(5-2t)\,dt = \left[\frac{1}{3}(5t-t^2)\right]_1^x$

For integral attempted with correct limits

M1

$= \frac{1}{3}(5x-x^2) - \frac{4}{3}$

For limits substituted in correct expression

A1

$F(x) = \frac{1}{3}(5x-x^2) - \frac{4}{3} + \frac{1}{3}$

F(1) added to give complete F(x)

A1

4

$= \frac{1}{3}(5x - x^2 - 3)$

AG

(ii) $\frac{1}{3}(5q - q^2 - 3) = 0.75$

Setting up equation

or

integral of $f(x)$ from q to $2 = 0.25$

M1

$4q^2 - 20q + 21 = 0$

or $q^2 - 5q + 5.25 = 0$

Reaching correct simplified quadratic

A1

$(2q - 3)(2q - 7) = 0$

or $q = 2.5 \pm 1$

Factorising for two solutions or using formula or calculator

m1

$q = 1.5$

Selecting only this one

A1

4

[15]

Q3.

(a) for $-5 \leq x \leq 15$

$$f(x) = \frac{d}{dx}F(x) = \frac{d}{dx}\left(\frac{x+5}{20}\right) = \frac{1}{20}$$

AG

B1

1

(b) (i) $P(X \geq 7) = 1 - F(7) = 1 - \frac{12}{20} = \frac{2}{5}$ or $\left[\frac{8}{20};\ \frac{4}{10};\ 0.4\right]$

Alternative:

Use of $f(x) = \frac{1}{20}$ ***or*** *graph* $\Rightarrow$

$P(X \geq 7) = \frac{1}{20} \times (15 - 7) = \frac{2}{5}$ *(oe)*

B1

1

(ii) $P(X \neq 7) = 1$

cao

B1

1

(iii) $E(X) = \frac{1}{2}(-5 + 15) = 5$

Alternative:

$E(X) = \int_{-5}^{15} \frac{x}{20}\,dx = \left[\frac{x^2}{40}\right]_{-5}^{15} = \frac{1}{40}$ *(225 − 25)* = $\frac{1}{40}$ *× 200; = 5*

***B1** (cao)*

B1

1

(iv) $E(3X^2) = \left. \int_{-5}^{15} \frac{3x^2}{20}\,dx \right\}$ (ignore limits)

M1

$$\left. \begin{array}{l} \left[\frac{x^3}{20}\right]_{-5}^{15} \\ \frac{1}{20}(3375+125) \\ 168\frac{3}{4}+6\frac{1}{4} \end{array} \right\}$$

correct limits seen / used

A1

= 175

(cao) (allow 174.9)

A1

3

Alternative:

$\text{Var}(X) = \frac{1}{12}(15 - -5)^2 = \frac{400}{12}$ (oe)

$E(3X^2) = 3\,E(X^2)$

(B1)

$E(3X^2) = 3 \times \left[\frac{400}{12} + 5^2\right]$

= 3 × $\left[\{\text{their Var}(X) > 0\} + \{\text{their E}(X)\}^2\right]$ ***used** (⇒ M1)*

(M1)

= 175

(A1)

(3)

[7]

Q4.

(a) $$F(x) = \int \frac{3}{8}(x^2 + 1)\mathrm{d}x$$

Ignore limits

M1

$$= \frac{3}{8}\left[\frac{x^3}{3} + x\right] \text{ or } = \frac{1}{8}x^3 + \frac{3}{8}x$$

Either

A1

$$= \frac{1}{8}x(x^3 + 3)$$

(including use of correct limits 0 and x ***or*** *+c used and evaluated) (AG)*

A1

3

(b) $$F(m) = \frac{1}{2}$$

B1

$$F(1) = \frac{1}{8} \times 1 \times 4 = \frac{1}{2}$$

AG

B1

2

(c) Upper quartile lies in range 1 < x < 2 such that $F(q) = \frac{3}{4}$

$$\frac{1}{2} + \int_1^q \frac{1}{4}(5 - 2x)dx = \frac{3}{4}$$

$$\int_1^q \frac{1}{4}(5 - 2x)dx = \frac{1}{4}$$

Alternative: $\int_q^2 \frac{1}{4}(5 - 2x)\mathrm{d}x = \frac{1}{4}$

M1

$$\left[5x - x^2\right]_1^q = 1$$

$$\left[5x - x^2\right]_q^2 = 1$$

$5q - q^2 - 4 = 1$

(10 – 4) – (5q – q^2) = 1

6 – 5q + q^2 = 1

$q^2 - 5q + 5 = 0$

$q^2 - 5q + 5 = 0$

A1

$$q = \frac{5 \pm \sqrt{25-20}}{2} \quad \text{or} \quad \frac{1}{2}\left(5 \pm \sqrt{5}\right)$$

Correct use of formula (OE) to give the two surd answers to given quadratic equation

M1

but 1 q 2 [or (q 2)]

m1

$$\therefore\ q = \frac{1}{2}\left(5 - \sqrt{5}\right)$$

Must qualify with a numerical comparison, not just quote the given answer; dep on M1; AG

A1 5

(d) $$\text{P}(X > 1.5) = \frac{1}{2}\left[\frac{1}{2} + \frac{1}{4}\right] \times \frac{1}{2}$$

$$\text{P}(X < 1.5) = 0.5 + \frac{1}{2}\left[\frac{3}{4} + \frac{1}{2}\right] \times \frac{1}{2} \quad \text{(M1)}$$

M1

$$= \frac{3}{16}(0.1875)$$

$$= \frac{1}{2} + \frac{1}{2} \times \frac{5}{4} \times \frac{1}{2}$$

$$= \frac{1}{2} + \frac{5}{16} = \frac{13}{16} \qquad \text{(A1)}$$

A1

$$\text{P}(X > q) = \frac{1}{4}(0.25)$$

$$\text{P}(X < q) = \frac{3}{4}(0.75) \qquad \text{(B1)}$$

B1

$$\text{P}(q < X < 1.5) = \frac{1}{4} - \frac{3}{16}$$

$$= \frac{1}{16} \quad (0.0625)$$

$$\text{P}(q < X < 1.5) = \frac{13}{16} - \frac{3}{4} = \frac{1}{16} \qquad \text{(A1)}$$

(0.0625)

A1

OR

$$\int_{1\frac{1}{2}}^{2} \frac{1}{4}(5-2x)\mathrm{d}x = \frac{3}{16} \text{ etc } \quad \text{(M1A1)}$$

OR

$$\int_{q}^{1.5} \frac{1}{4}(5-2x)\mathrm{d}x = \frac{1}{4}\left[5x - x^2\right]_q^{1.5} \quad \text{(M1)}$$

(correct integration and limits)

Allow use of q = 1.38 to q = 1.382 in limits for M1

*Whatever follows **must** be **exact***

$$= \frac{1}{4}\left[(7.5 - 2.25) - \left(5q - q^2\right)\right] \quad \text{(A1)}$$

*for use of $5q - q^2 = 5$ **or** showing*

$5q - q^2 = 5$ by substituting $q = \frac{1}{2}\left(5 - \sqrt{5}\right)$ *(A1)*

$$= \frac{1}{4}[5.25 - 5] = \frac{1}{16} \quad \text{(A1)}$$

NB statement $\mathrm{F}(1.5) - \frac{3}{4} = \frac{1}{16}$ (OE) scores 4 marks

4

Alternative:

$$\int_{q}^{1.5} \frac{1}{4}(5-2x)\mathrm{d}x = \left[-\frac{1}{16}(5-2x)^2\right]_{\frac{5-\sqrt{5}}{2}}^{1.5} \quad \text{(M1)}$$

$$= -\frac{1}{16}(4) - \left[-\frac{1}{16}\left(\sqrt{5}\right)^2\right] \quad \text{(sub)} \quad \text{(A1)}$$

$$= -\frac{4}{16} + \frac{5}{16} \quad \text{(A1)}$$

$$= \frac{1}{16} \quad \text{(A1)}$$

Alternative using F(x):

for $1 \le x \le 2$

$$\mathrm{F}(x) = \frac{1}{2} + \int_{1}^{x} \frac{1}{4}(5-2x)dx$$

$$= \frac{1}{2} + \frac{1}{4}\left[5x - x^2\right]_1^x$$

$$= \frac{1}{2} + \frac{1}{4}\left[\left(5x - x^2\right) - (5-1)\right]$$

$$= \frac{1}{4}\left(2 + 5x - x^2 - 4\right)$$

$$= \frac{1}{4}\left(5x - x^2 - 2\right) \text{ (seen or used)} \quad \text{(M1)}$$

$$F(1.5) = \frac{1}{4}(7.5 - 2.25 - 2) = \frac{3.25}{4}$$

$$= 0.8125 = \frac{13}{16} \quad \text{(A1)}$$

$$F(q) = \frac{1}{16}\left(50 - 10\sqrt{5} - \left(25 - 10\sqrt{5} + 5\right) - 8\right)$$

$$= \frac{12}{16} \text{ OE} \quad \text{(B1)}$$

$$P(q < X < 1.5) = \frac{13}{16} - \frac{12}{16} = \frac{1}{16} \quad \text{(A1)}$$

[14]

Q5.

(a) Median = 1

B1

Lower quartile = $\frac{1}{2}$

B1

2

(b) F(1) = $\frac{1}{2}$

For $1 \le x \le 4$

$\int \frac{1}{18}(x-4)^2 \, dx$

ignore limits

M1

$$= \left[\frac{1}{54}(x-4)^3\right]_1^x$$

$$= \left[\frac{1}{54}(x-4)^3 + \frac{1}{2}\right]$$

Correct integration + correct limits seen or used

A1

$$F(x) = \left[\frac{1}{54}(x-4)^3 + \frac{1}{2}\right] + \frac{1}{2}$$

adding $\frac{1}{2}$ ***or*** *F(1)*

m1

$= 1 + \frac{1}{54}(x-4)^3$

CAO (AG)

A1

Alternative

$\int \frac{1}{18}(x-4)^2\,dx = \frac{1}{54}(x-4)^3 + c$

(M1)

F(1) = $\frac{1}{2}$ $\Rightarrow$ $c = 1$

(m1)(A1)

F(x) = 1 + $\frac{1}{54}$ $(x-4)^3$

(A1) **4**

Alternative

$\int \frac{1}{18}(x-4)^2\,dx$ (M1)

$= \int_1^x \frac{1}{18}(x^2 - 8x + 16)\,dx$

$= \frac{1}{18}\left[\frac{x^3}{3} - 4x^2 + 16x\right]_1^x$ (A1)

$F(x) = \frac{1}{2} + \frac{1}{54}[x^3 - 12x^2 + 48x]_1^x$ (m1)

$= \frac{1}{2} + \frac{1}{54}(x^3 - 12x^2 + 48x - 37)$

$= 1 + \frac{1}{54}(x^3 - 12x^2 + 48x - 64)$

$= 1 + \frac{1}{54}(x-4)^3$ (A1)

(c) P(2 ≤ X ≤ 3) = $\frac{53}{54} - \frac{46}{54}$

F(3) – F(2)

M1

$= \frac{7}{54}$ (0.130)

$0.1\dot{2}9\dot{6}$

A1 2

(d) (i) $F(q) = \frac{3}{4}$

use *of F(q) =* $\frac{3}{4}$

M1

$1 + \frac{1}{54}(q-4)^3 = \frac{3}{4}$

$\frac{1}{54}(q-4)^3 = -\frac{1}{4}$

(either)

M1

(×54) $\Rightarrow (q - 4)^3 = -13.5$

AG

A1 3

(ii) $q - 4 = \sqrt[3]{-13.5} = -2.3811 \quad q = 1.619\ (3\text{dp})$

CAO

B1 1

[12]

Q6.

(a) $P\left(-\frac{3c}{4} < X < \frac{3c}{4}\right)$

$= \frac{\frac{3c}{4} + c}{4c} - \frac{\frac{-3c}{4} + c}{4c}$

or = $\frac{3c}{2} \times \frac{1}{4c}$

M1

$= \frac{6c}{16c}$

$= \frac{3}{8}$ or 0.375

CAO

A1 2

(b) For $-c \le x \le 3c$

$$f(x) = \frac{d}{dx}\left(\frac{x+c}{4c}\right)$$

use of f(x) = F′(x)

M1

$$= \frac{1}{4c}$$

For $x > 3c$ and $x < -c$

$$f(x) = \frac{d}{dx}(F) = 0$$

for $\frac{1}{4c}$ *and 0*

A1

2

(c) (i) Rectangular distribution:

$$E(X) = \frac{1}{2}(-c + 3c) = c$$

B1

1

(ii) Var $(X) = \frac{1}{12}(3c - -c)^2 = \frac{4c^2}{3}$

Allow $\frac{16c^2}{12}$

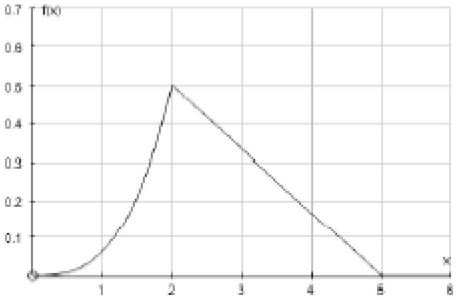
B1

1

[6]

Q7.

(a)



for concave curve from (0, 0) to (2, 0.5)

B1

for straight line from (2, 0.5) to (5, 0)

B1

for axes [2, 5; 0.5] seen

B1

3

(b) $P(X \geq 2) = \frac{1}{2} \times 3 \times 0.5 = 0.75$

$\Rightarrow$ F(2) = 0.25

Alternative:

$$\int \frac{1}{6}(5-x)\,dx = \frac{1}{6} \times \frac{(5-x)^2 \times (-1)}{2}$$

$$= \frac{1}{12}(5-x)^2$$

$2 \leq x \leq 5$

$$F(x) = F(2) + \int_2^x \frac{1}{6}(5-x)\,dx$$

$$= 0.25 + \frac{1}{6}\left[5x - \frac{x^2}{2}\right]_2^x$$

M1

$$= 0.25 + \frac{1}{6}\left(5x - \frac{x^2}{2}\right) - \frac{1}{6}(10-2)$$

A1

$$= 0.25 - \frac{8}{6} + \frac{5x}{6} - \frac{x^2}{12}$$

$$= -\frac{1}{12}(x^2 - 10x + 13)$$

M1

$$= 1 - \frac{1}{12}(5-x)^2$$

A1

Or

F(x) = 1 – Area Δ (base x, 5)

$= 1 - \frac{1}{2}(5-x)\frac{1}{6}(5-x)$

$= 1 - \frac{1}{12}(5-x)^2$

4

(c) $P(X \leq 4) = F(4)$

$= 1 - \frac{1}{12}(5-4)^2 = \frac{11}{12}$ (0.916 to 0.917)

B1

$F(3) = 1 - \frac{1}{12}(2)^2 = \frac{2}{3}$ (0.667)

B1

$P(X \geq 3 \text{ and } X \leq 4) = F(4) - F(3)$

$= \frac{11}{12} - \frac{2}{3} = \frac{1}{4}$ (0.25)

B1

$P(X \geq 3 | X \leq 4) = \frac{F(4) - F(3)}{F(4)}$

M1

$= \frac{1/4}{11/12} = \frac{3}{11}$

A1

Alternative:

$P(X \geq 3 \mid X \leq 4)$

$= \frac{F(4) - F(3)}{F(4)}$ (M1)

$= 1 - \frac{F(3)}{F(4)}$

$= 1 - \frac{2/3}{11/12}$

$= 1 - \frac{8}{11}$ (B1) $(0.7\dot{2}\dot{7})$

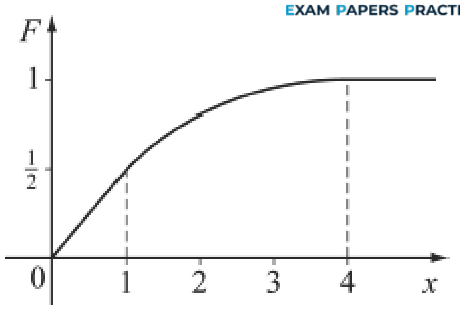
$= \frac{3}{11}$ *(AWFW 0.272 to 0.273)*

5

[12]

Q8.

(a) (i)



B1 for axes 0 to 4 & 0 to 1

B1 for straight line 0 – $\left(1, \frac{1}{2}\right)$

B1 for convex curve from $\left(1, \frac{1}{2}\right)$ *to (4,1)*

B1 for at least the straight line for x > 4

B4 **4**

(ii) $F(q_1) = 0.25$ $\Rightarrow \frac{1}{2}q_1 = 0.25$

From sketch, or from F(x), Median = 1.

M1

$\Rightarrow q_1 = \frac{1}{2}$

$\frac{1}{2}x$ *is linear on (0, 0) to* $\left(1, \frac{1}{2}\right)$

$\therefore q_1 = \frac{1}{2}$ ***AG***

A1 **2**

(iii) $F(1.6) = 0.744$
$F(1.7) = 0.775$

M1

$F(q_3) = 0.75$

M1

$\Rightarrow 1.6 < q_3 < 1.7$

AG

A1 **3**

(b) (i) $f(x) = F(x)$

M1

$$\Rightarrow f(x) = \frac{1}{2} \text{ for } 0 \le x \le 1$$

A1

$$\Rightarrow a = \frac{1}{2}$$

f(1) = α = 9β *B1*

$$\int_{-\infty}^{\infty} f(x)dx = 1 \Rightarrow$$

$$[ax]_0^1 + \left[\frac{\beta(x-4)^3}{3}\right]_1^4 = 1$$ *M1*

$\Rightarrow$ for $1 \le x \le 4$

$$f(x) = \frac{1}{54}(3x^2 - 24x + 48)$$

A1

$$= \frac{3}{54}(x^2 - 8x + 16)$$

M1

$$= \frac{1}{18}(x-4)^2$$

$$\Rightarrow \beta = \frac{1}{18}$$

A1

$\Rightarrow$ ***α + 9β = 1*** *A1*

Solving: *M1*

$a = \frac{1}{2}$ *and* $\beta = \frac{1}{18}$ *A1*

5

(ii)

$$E(X) = \int_0^1 \frac{1}{2}x\,dx + \int_1^4 \frac{1}{18}(x^3 - 8x^2 + 16x)dx$$

Both seen

M1

$$= \left[\frac{1}{4}x^2\right]_0^1 + \frac{1}{18}\left[\frac{x^4}{4} - \frac{8x^3}{3} + 8x^2\right]_1^4$$

A1A1

$= \frac{1}{4} + \frac{7}{8}$

Dependent on M1

m1

CAO

A1

$= 1\frac{1}{8}$

5

[19]

Q9.

(a) P(X < 0) = F(0)

M1

$= \frac{1}{k+1}$

A1

2

(b) (i) f(x) = $\frac{d}{dx}\left(F(x)\right)$

use of

M1

$= \frac{1}{k+1} \times \frac{d}{dx}(x+1)$

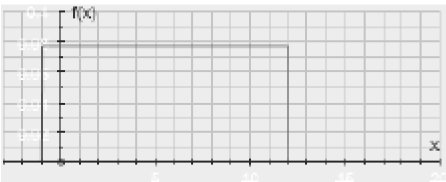
$= \frac{1}{k+1}$ for $-1 \le x \le k$

AG; $\frac{1}{k+1}$ *clearly deduced*

A1

2

(ii)



B2

2

(c) $\frac{1}{12}(k+1)^2 = \left(\frac{1}{2}(k-1)\right)^2$

M1

$\frac{1}{12}(k+1)^2 = \frac{1}{4}(k-1)^2$

$(k+1)^2 = 3(k-1)^2$

$k^2 + 2k + 1 = 3k^2 - 6k + 3$

M1

$2k^2 - 8k + 2 = 0$

$k^2 - 4k + 1 = 0$

A1

$k = \frac{4 \pm \sqrt{16-4}}{2}$

m1

$k = 2 - \sqrt{3}$ or $k = 2 + \sqrt{3}$

AG

A1

Alternative:

$(k+1)^2 = \left(\sqrt{3}(k-1)\right)^2$

$\left[(k+1) + \sqrt{3}(k-1)\right]\left[(k+1) - \sqrt{3}(k-1)\right] = 0$

$k = \frac{\sqrt{3}-1}{1+\sqrt{3}}$ or $\frac{1+\sqrt{3}}{\sqrt{3}-1}$

$k = 2 - \sqrt{3}$ or $k = 2 + \sqrt{3}$

5

[11]

Q10.

(a) P(X < 0) = F(0)

M1

$= \frac{1}{k+1}$

A1

2

(b) $(q_1 + 1) \times \frac{1}{k+1} = \frac{1}{4}$

alternative (from a sketch)

M1

$q_1 + 1 = \frac{1}{4}(k+1)$

A1

$q_1 = \frac{1}{4}(k+1) - 1$

OE

A1

3

(c) $f(x) = \frac{d}{dx}(F(x))$

use of

M1

$$\left.\begin{array}{ll} = \frac{1}{k+1} \times \frac{d}{dx}(x+1) & \\ = \frac{1}{k+1} & -1 \le x \le k \\ = 0 & \text{otherwise} \end{array}\right\}$$

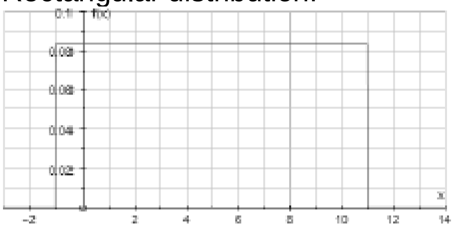
AG; $\frac{1}{k+1}$ *clearly deduced*

A1

2

(d) (i) $k = 11 \Rightarrow f(x) = \begin{cases} \frac{1}{12} & -1 \le x \le 11 \\ 0 & \text{otherwise} \end{cases}$

Rectangular distribution:



horizontal line on [–1, 11]

B1

at $f = \frac{1}{12}$

B1

2

(ii) $E(X) = \frac{1}{2}(-1 + 11) = 5$

B1

$Var(X) = \frac{1}{12}(11 - -1)^2 = 12$

B1

2

(iii) $P(q_1 < X < E(X)) = P(2 < X < 5)$

$= (5 - 2) \times \frac{1}{12}$

M1

= 0.25

AG

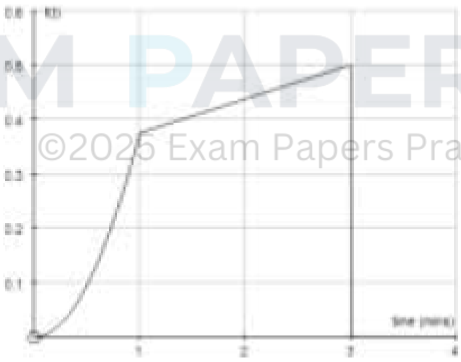
A1

2

[13]

Q11.

(a)



for curve

B1

for line

B1

for axes

B1

3

(b) $P(T \geq 1) = \frac{1}{2} \times \frac{7}{8} \times 2 = \frac{7}{8}$

OE

M1A1

2

(c) (i) For $1 \le t \le 3$

$$\int_1^t \frac{1}{16}(t+5)\,dt = \left[\frac{1}{32}t^2 + \frac{5}{16}t\right]_1^t$$

M1A1

$$F(1) = \frac{1}{8}$$

B1

$$F(t) = \frac{1}{8} + \frac{1}{32}t^2 + \frac{5}{16}t - \frac{11}{32}$$

Use of : $F(t) = F(1) + \int_1^t \frac{1}{16}(t+5)\,dt$

M1

$$F(t) = \frac{1}{32}\left(t^2 + 10t - 7\right)$$

AG

A1

5

Alternative:

$$\int \frac{1}{16}(t+5)\,dt$$

$$= \frac{1}{16}\left(\frac{1}{2}t^2 + 5t + c\right)$$

(M1)(A1)

$$F(1) = \frac{1}{8}$$

(B1)

$$\Rightarrow c = -3.5$$

(M1)

$$F(t) = \frac{1}{32}(t^2 + 10t - 7)$$

(A1)

(ii) $\frac{1}{32}\left(m^2 + 10m - 7\right) = 0.5$

M1

$m^2 + 10m - 23 = 0$

A1

$m = \frac{-10 \pm \sqrt{192}}{2} = -5 \pm \sqrt{48}$

(or any valid method)

m1

$= -5 \pm 4\sqrt{3}$

$(m > 0)$

$m = 4\sqrt{3} - 5 = 1.93$

(1.9282)

A1 **4**

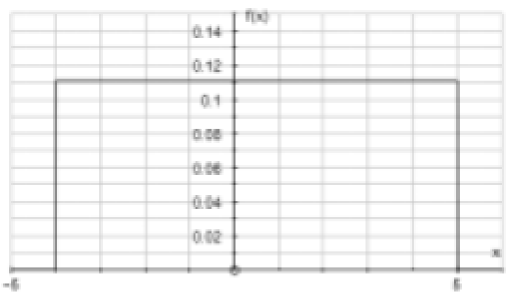
[14]

Q12.

(a) $f(x) = \begin{cases} \frac{1}{9} & -4 \le x \le 5 \\ 0 & \text{otherwise} \end{cases}$

M1A1 **2**

(b)



horizontal line from –4 to 5

B1

for drawn at $\frac{1}{9}$

B1 **2**

(c) $P(X > 2) = \frac{1}{9} \times 3$

F(5) – F(2)

$= 1 - \frac{2}{3}$

M1

$= \frac{1}{3}$

A1

2

(d) Mean = $\frac{1}{2}$

B1

Variance = $\frac{1}{12} \times 81$

= 6.75

B1

2

[8]

Q13.

(a) E(*T*) = $\int_0^1 t\,f(t)\,dt$

$= \int_0^1 4t^2(1-t^2)\,dt$

M1

$= \left(\frac{4t^3}{3} - \frac{4t^5}{5}\right)\Big|_0^1$

A1

$= \frac{4}{3} - \frac{4}{5}$

A1

$= \frac{8}{15}$

AG

3

(b) (i) $F(t) = P(T < t) = \int_0^t f(t)\,dt$

$\int_0^t 4t(1 - t^2)dt$

M1

$= (2t^2 - t^4)\Big|_0^t$

$= 2t^2 - t^4$

A1

2

(ii) $P(\mu < T < m) = F(m) - F(\mu)$

$\Downarrow$

M1

$F(m) = 0.5$

B1

$F(\mu) = F\left(\frac{8}{15}\right) = 0.4880$

B1

$P(\mu < T < m) = 0.5 - 0.488$

0.5–their F(μ)

M1ft

$\therefore = 0.012$

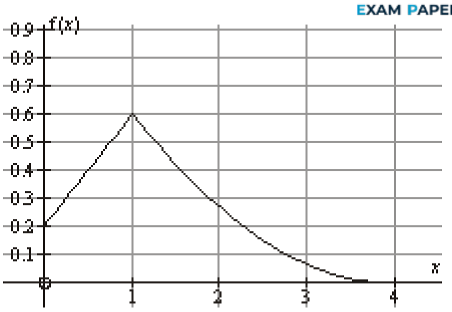
A1

5

[10]

Q14.

(a)



B1 for line segment
(0, 0.2) to (1, 0.6)
B1 for correct shaped curve
(1, 0.6) to (4 ,0)

B2

2

(b) (i) for $0 \le x \le 1$

$$F(x) = \int_0^x \frac{1}{5}(2x)\,dx$$

M1

$$= \frac{1}{5}(x^2 + x)\Big|_0^x$$

A1

$$\frac{1}{5}x(x+1)$$

A1

3

(ii) $P(X \ge 0.5) = 1 - F(0.5)$

$$= 1 - \frac{1}{5} \times 0.5 \times 1.5$$

M1

$$= 1 - 0.15$$
$$= 0.85$$

(OE)

A1

2

(iii) $F(q_1) = 0.25$

$$\frac{1}{5}x(x+1) = 0.25$$

↓

B1

$F(0.5) = \frac{3}{20} = 0.15 < 0.25$

$F(0.75) = \frac{21}{80} = 0.2625 > 0.25$

Attempt to solve

↓

$q_1 = 0.725$

⇓

M1

$\therefore \quad 0.5 < q_1 < 0.75$

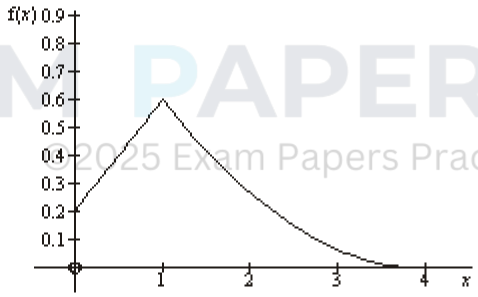
$0.5 < q_1 < 0.75$ *AG*

A1 3

[10]

Q15.

(a)



B1 for line segment (0,0.2) to (1,0.6)
B1 for correctly shaped curve (1,0.6) to (4,0)

B2 2

(b) (i) for $0 \le x \le 1$

$F(x) = \int_0^x \frac{1}{5}(2x+1)dx$

Ignore limits

M1

$= \left[\frac{1}{5}(x^2 + x)\right]_0^x$

Ignore limits

A1

$= \frac{1}{5}x(x+1)$

A1

3

(ii) $P(X \le 1) = F(1)$

$= \frac{2}{5}$

B1

1

(iii) $P(X \ge x) = \frac{17}{20} \Rightarrow F(x) = \frac{3}{20}$

M1

$\frac{1}{5}x(x+1) = \frac{3}{20}$

m1

$x(x+1) = \frac{3}{4}$

$x^2 + x - \frac{3}{4} = 0$

A1

$\left(x - \frac{1}{2}\right)\left(x + \frac{3}{2}\right) = 0$

Any valid method attempted

m1

$x - \frac{1}{2}$

A1

5

(iv) Since $F(1) = 0.4$, q lies in $0 \le r \le 1$

F(q) = $\frac{1}{5}(q^2 + q) = 0.25$

M1

$\Rightarrow \quad q^2 + q = 1.25$

$q^2 + q - 1.25 = 0$

A1

$$\Rightarrow \quad q = \frac{-1 \pm \sqrt{1 - 4 \times (-1.25)}}{2}$$

m1

$q = \frac{1}{2}\left(\sqrt{6} - 1\right) \quad (q > 0)$

AWFW (0.724 to 0.725)

A1

4

[15]

Q16.

(a) P(T = 0) = 0

B1

1

(b) (i) P(T < 2) = 1 – $e^{\frac{-2}{4}}$ = $1 - e^{-0.5}$

M1

= 0.393
AWRT

A1

2

(ii) P(2 ≤ T < 6) = F (6) – F (2)

M1

= $e^{-0.5} - e^{-1.5}$
PI

m1

= 0.383

0.383 to 0.384

A1

3

(c) We require P(T > 4 | T > 1)

M1

$$= \frac{P(T > 4 \text{ and } T > 1)}{P(T > 1)}$$

m1

$$= \frac{e^{-1}}{e^{-0.25}}$$

m1

= 0.472

CAO

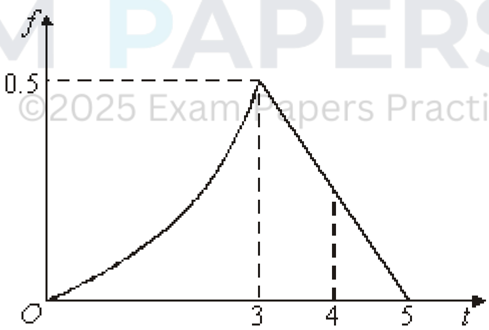
A1

4

[10]

Q17.

(a)



Curve from(0, 0) to (3, 0.5)

B1

Straight line from (3, 0.5) to (5, 0)

B1

2

(b)

F(t) = $0 \le t \le 3$

$3 \le t \le 5$

$$\begin{cases} \frac{t^3}{54} \\ \frac{1}{8}(10t = t^2 - 17) \end{cases}$$

B1

M1M1

A1

4

(c) P(T < 4) = F(4) = $\frac{1}{8}$(40 – 16 – 17)

M1

$\frac{7}{8}$ or 0.875

Alternative (c): *1 –* $\frac{1}{2}$ × *1* × f*(4)*

1 – $\frac{1}{2}$ × *1* × $\frac{1}{4}$

1 – $\frac{1}{8}$ = $\frac{7}{8}$

A1

2

[8]

Q18.

(a) (i) f(t) = $2e^{-2t}$, ($t \geq 0$)

no penalty for omitting $t \geq 0$

B1

1

(ii) Mean = 0.5

CAO

B1

Standard deviation = 0.5

CAO

B1

2

(b) P(T > 3) = 1 – F(3)

= 1 – (1 – e^{-6})

or by integration

M1

= 0.0248

AWRT 0.00248

A1

2

(c) Median m satisfies F(m) = 0.5

$1 - e^{-2m} = 0.5$

or by integration

M1

$e^{-2m} = 0.5$

$-2m = \ln(0.5)$

or $e^{2m} = 2$

$m = -0.5 \ln(0.5)$ A1

giving 0.5 ln (2)

(= 0.347)

Median delay = 20.82

= 21 seconds to nearest second

CAO

A1

3

[8]

Examiner reports

Q1.

In attempting this question, the population of students split essentially into two parts. The major part understood the concept of F(t) and its relationship to f(t) and generally scored very highly. The minor part usually started by integrating F(t) and thereafter gained very few marks. Amongst the better students, it was disappointing that so many failed to convert 5.79 weeks into days, as required, and equally that so many at this level differentiated t^3 and obtained $2t^2$.

Q2.

The sketch was reasonably well done and most students correctly showed the integration from 0 to x to find F(x). The calculation of the lower quartile was also well done. In part (c)(i), there was much fudging of the answer after the integral from 1 to x gave −4 while the required answer was −3, causing many to lose the mark for the correctly evaluated integral when they spuriously altered it. Often an incorrect answer to part (i) was carried over into part (ii) giving an incorrect equation, with the given F(x) been ignored. It was disconcerting to see how many students at this level could not even recognise or subsequently solve a quadratic equation. On the other hand, many students scored highly on all parts of this question.

Q3.

In part (a), most students were able to indicate that the differentiation of F*(x*) was required even though use of the correct notation was often sadly lacking. In part (b), those who realised that they were dealing with a continuous *rectangular* distribution, found parts (i), (ii) and (iii) very easy; simply writing down the required answers. Unfortunately, in part (b)(i), some students thought that the answer was 0.6 whilst others assumed a discrete distribution and consequently attempted to find 1 ꝍ P (X □ 6) . In part (b)(ii), a common incorrect answer was P (X □ 7) = 0. In part (b)(iii), the correct answer E (X) = 5 was usually seen but most students, unnecessarily, used calculus to evaluate $\int_{-5}^{15} \frac{x}{20}$ thus wasting time. In part (b)(iv), the great majority chose to correctly evaluate $\int_{-5}^{15} \frac{3x^2}{20}$ to achieve the required answer of 175.

Q4.

Although most candidates managed to correctly integrate $\frac{3}{8}(x^2+1)$ in part (a), many then failed to use the correct limits required to find F(x), with the vast majority failing to consider any limits at all but simply stating the given answer.

In part (b), many candidates stated F (m) = 0.5 but then failed to verify that m = 1 satisfied this condition. Part (c) was not well answered except by the more able candidates. Many started from the incorrect assumption that $\int_0^q \frac{1}{4}(5-2x)\mathrm{d}x = \frac{3}{4}$. Consequently, the given equation was often quoted without any foundation at the end of a lot of spurious work.

It was also very disappointing to see candidates on this paper unable to solve the given

quadratic equation. Those who did manage to obtain two surd answers were often then unable to give an adequate numerical comparison in order to justify the selection of the given answer. Most candidates simply wrote down the given answer without any justification whatsoever. This lost them the final two marks for this part of the question. Part (d), although relatively straightforward, confounded many candidates as very few realised that all they had to do was to evaluate
F (1.5) – F(q) where F(q) = ¾ and F(1.5) = 13/16.

Q5.

Part (a) was answered very poorly by too many candidates. Candidates were asked to **state** values for the median and lower quartile of X. This should have indicated to candidates that the answers required little work, but this was often not picked up. Some candidates obtained the correct values but these were often invalidated by incorrect methods. Those candidates who employed calculus often had more success, as they realised that F(m) = 0.5 and F(q_1) = 0.25 were required.

Part (b) was often done well by those candidates who realised that F(x) = 1/54(x – 4)^3 + c, as they usually went on to use either F(1) = 1/2 or F(4) = 1 to show that c = 1. Some candidates, having been given the answer in the question, simply wrote this down at the end of what was usually a totally incorrect method.

Part (c) was usually done by correctly attempting to evaluate F(3) – F(2) but the arithmetic then defeated many.

Most candidates realised that F(q) = 3/4 was required and consequently went on to gain full marks for part (d)(i). The correct answer to part (d)(ii) was often seen, even by those candidates who could not do part (d)(i). However, some failed to give the answer to three decimal places and so lost the mark.

Q6.

On the whole, attempts at this question were rather disappointing. In part (a), when candidates did demonstrate that they knew that $F\left(\frac{3c}{4}\right) - F\left(-\frac{3c}{4}\right)$ was required, the algebraic manipulation then often defeated them. There were very few candidates who realised that the given F(x) was the cumulative distribution function for a *rectangular distribution*. Those who did recognise this fact were simply able to find the area of the rectangle of length $\frac{3c}{2}$ and height $\frac{1}{4c}$ to give the correct answer of $\frac{3}{8}$.

In part (b), although many candidates realised that they had to use the relationship f(x) = F′(x), they were then either unable to differentiate $\frac{x+c}{4c}$ with respect to x to give $\frac{1}{4c}$ and/or neglected to explain what happens when they differentiate F(x) in the two regions $x < -c$ and $x > 3c$, simply stating the given answer did not suffice.

In part (c), although many candidates realised at this stage that they were dealing with a rectangular distribution and so used the appropriate formulae for E(X) and Var(X) from the table provided on page 11 of the booklet to obtain the required answers, many others wasted a lot of time on often fruitless attempts at integration. Also, candidates' inability to simplify $E(X) = \frac{1}{2}(2c)$ and $Var(X) = \frac{1}{12}(4c)^2$ resulted in a loss of marks.

Q7.

A number of candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from their graphics calculator display. All such attempts lost some or all of the marks available. The axes should ideally have been drawn with the aid of a ruler and pencil and should have contained the appropriate scales; $O(0, 0)$, (2, 0) and (5, 0) on the x-axis and (0, 0.5) on the y-axis as a bare minimum. A concave curve from $O(0, 0)$ to (2, 0.5) and a straight line from (2, 0.5) to (5, 0) should then have been drawn.

Most candidates realised that $\int_2^x \frac{1}{6}(5-x)\mathrm{d}x$ was required somewhere in part (b) even though some were unsure of the limits and what this integral gave them. Some thought incorrectly that $F(x) = \int \frac{1}{6}(5-x)\mathrm{d}x \Rightarrow F(x) = -\frac{1}{12}(5-x)2^2 + c$, followed by $F(5) = 1 \Rightarrow c = 1$, and hence that $F(x) = 1 - \frac{1}{12}(5-x)^2$ as required. This was very plausible but, unfortunately, the original assumption that $F(x) = \int \frac{1}{6}(5-x)\mathrm{d}x$ was false! However, that said, there were some excellent and fully correct solutions to part (b) where the correct expression $F(x) = F(2) + \int_2^x \frac{1}{6}(5-x)\mathrm{d}x$ or $F(x) = \int_2^x \frac{1}{6}(5-x)\mathrm{d}x + k$ was used.

In part (c), the great majority of candidates were able to show that $F(4) = \frac{11}{12}$ and $F(3) = \frac{2}{3}$, and then went on to show that $P(3 \le X \le 4) = \frac{1}{4}$. However, some candidates treated this as if it were a discrete distribution, stating that $P(X \ge 3) = 1 - P(X \le 2)$. Unfortunately, all but the very best candidates assumed incorrectly that $P(X \ge 3 \mid X \le 4) = P(3 \le X \le 4) = \frac{1}{4}$. Conditional probability was required using the fact that $P(X \ge 3 \mid X \le 4) = 1 - \frac{F(3)}{F(4)} = 1 - \frac{8}{11} = \frac{3}{11}$ or $P(X \ge 3 \mid X \le 4) = \frac{F(4) - F(3)}{F(4)} = \frac{1/4}{11/12} = \frac{3}{11}$.

Q8.

In general, the attempts at this question were very poor. It should be emphasised, yet again, that when an answer is given in a question, candidates must take extra care to explain what they are doing. This was often not seen in answers to parts (a)(ii), (a)(iii) and (b)(i), where many candidates showed insufficient working to achieve full credit.

Part (a)(i) asked candidates to sketch the graph of F. Many interpreted this as allowing the drawing of a rather scruffy freehand diagram without a scale. This was not what was intended. It was expected that scaled (approximately) and labelled axes be drawn (using a ruler and preferably a pencil) and care be taken in drawing the curved part of the sketch.

Although most candidates attempted the straight line from the origin to (1, ½) correctly, there was much less success with the curve from (1, ½) to (4, 1). The part of the graph defined by F(*x*) = 1 was often not drawn, and the part defined by F(*x*) = 0 was rarely seen at all.

Part (a)(ii) was not explained well by many candidates. Many thought incorrectly that dividing 4 (the length of the interval over which most of them drew their sketch) by 4 would give them the value of the lower quartile, whilst others assumed that (the value of the lower quartile) = $\frac{1}{2}$ × the value of the median) without any explanation as to why this was the case for this distribution.

It was, however, pleasing to see good candidates state correctly that $F(q_1) = \frac{1}{2}q_1 = 0.25$, leading to the given answer of $q_1 = \frac{1}{2}$. A variety of methods, mostly correct, were employed in part (a)(iii). The vast majority of candidates realised that the upper quartile was in the interval [1, 4] and used the fact that $F(q_3) = 0.75$ to good effect. Most used different rearrangements of the resultant cubic equation, whilst others used their calculators to show that $q_3 = 1.62$ (3 sf), to achieve the required given result.

The attempts at part (b)(i) were varied and mostly disappointing, with many candidates using the result $\alpha = \frac{1}{2}$ to show that $\beta = \frac{1}{18}$ without firstly explaining why they thought that $\alpha = \frac{1}{2}$. Some found only one relationship between α and β (usually either $\alpha = 9\beta$ or $\alpha + 9\beta = 1$) and then simply substituted the given values of α and β to show that these values did work. This was not felt to be sufficient to gain full credit since each of these equations, taken in isolation, had infinitely many solutions, of which the given solution was just one such pair of values. However, there were a few excellent solutions, the most successful of these first showing that on the interval [0, 1], $f(x) = \alpha = F'(x) = \frac{1}{2}$. This was then followed by use of the property that f was continuous at $x = 1$ to give $f(1) = \alpha = 9\beta$, thus giving $\beta = \frac{1}{18}$ as required.

Part (b)(ii) was a good source of marks, even for those candidates who gained virtually no credit on earlier parts of the question. These candidates showed good algebraic and integration skills in finding correctly that E(*X*) = 1.125. A small minority of candidates, instead of using any integration methods, used their calculators to find the correct numerical answer, usually after stating correctly that E(*X*) = $\int_0^1 \frac{1}{2}x\,dx + \int_1^4 \frac{1}{18}x(x-4)^2\,dx$. This method gained the full 5 marks for those candidates who stated the correct numerical value but only 1 mark for those who did not. Whilst the use of calculators is encouraged in data analysis and in the evaluation of numerical expressions, candidates run the risk of losing marks if they acquire an incorrect numerical answer without any intermediate working.

Q9.

Although there was one fully correct solution seen to this question, the majority of candidates could not cope with part (c). Consequently this was the worst answered question on the paper. In part (a), F(0) = $\frac{1}{k+1}$ was all that was required but this seemed to be beyond some candidates. In part (b)(i), candidates attempted to use the relationship f(x) = $\frac{d}{dx}(F(x))$ in order to gain the required given answer. However, attempts at sketching the graph of f in part (b)(ii) were disappointing.

Q10.

In part (a), F(0) = $\frac{1}{k+1}$ was all that was required, but the use of F(0) – F(–1) = $\frac{1}{k+1}$ was not penalised. In part (b), most candidates realised that they needed to use the fact that F(q_1) = 0.25 and so went on to determine the correct expression for q_1. Unfortunately many candidates must have thought that k was the lower quartile since they found an expression for k in terms of x. In part (c), candidates were expected to use the relationship f(x) = $\frac{d}{dx}(F(x))$ in order to gain the required given answer. Unfortunately, attempts at differentiating F(x) = $\frac{x+1}{k+1}$ were, in the main, unconvincing with little or no attention paid to F(x) = 0 and F(x) = 1. In part (d)(i), most candidates gained full marks. Many candidates used integration methods in part (d)(ii) which, although correctly done, did waste a lot of time, especially as the answer for E(X) = 5 could have been written down by simply considering the sketch drawn in part (d)(i). Similarly, Var(X) = $\frac{1}{12}(11 - -1)^2$ was a more efficient method of finding the variance. There were many well-explained solutions to part (d)(iii) with most candidates realising that P(2 < X < 5) was required. Many candidates correctly stated that P(X < q_1) = 0.25 but then assumed that P(X < E(X)) = 0.5 without any explanation. This statement was true here since, for a rectangular distribution, the median and the mean take the same value.

Q11.

Most candidates, in part (a), made a good attempt at sketching the graph of the probability function, although there were the usual weak attempts where candidates did not recognise that f(t) = $\frac{3}{8}t^2$ is a quadratic curve whilst f(t) = $\frac{1}{16}(t+5)$ is a straight line.

Part (b) was rarely attempted using the most efficient method of finding the area of a trapezium, but those candidates who did usually did so correctly. The vast majority attempted to use integration when answering this part of the question, with either $\int_1^3 \frac{1}{16}(t+5)\,dt$ or $1 - \int_0^1 \frac{3}{8}t^2\,dt$ often being used correctly. However, integration was not necessary and its use here again used up valuable time which could have been helpful to some candidates elsewhere. It was disappointing to see that only the most able students coped well with part (c)(i). Very few realised that F(t) = F(1) + $\int_1^t f(t)\,dt$ was required here, with the vast majority simply evaluating $\int_1^t \frac{1}{16}(t+5)\,dt$ and then writing down the answer that was given in the question. Part (c)(ii) was tackled much better, with many candidates obtaining the correct answer. The negative root to the obtained quadratic equation was correctly rejected but often without any justification being given.

Q12.

There were very many excellent solutions to this question with full marks seen on a regular basis; however some candidates seem to struggle with questions where the cumulative distribution function is given, rather than the probability density function. In part (a), some candidates were unable to differentiate the function and some treated the given function as the probability density function and attempted, usually with very little success, some sort of integration. Of the very many candidates who did manage to differentiate F(x) correctly, a few failed to quote the full answer for f(x), simply giving the answer as $\frac{1}{9}$.

In part (b), those candidates who had obtained the correct answer to part (a) usually drew the correct graph. Unfortunately, some candidates showed their lack of understanding of what was given in the question by simply attempting to draw the graph of F(x). Part (c) was done well by those candidates who understood the question, many using the graph that they had drawn in part (b) to help them to obtain the correct answer. For those who realised that this was a continuous rectangular distribution, part (d) was done very quickly by using the correct formulae from page 11 of the supplied booklet of formulae and statistical tables. However, some candidates attempted, usually with little success, to use integration methods in order to find the mean and variance of the given distribution.

Q13.

Candidates seemed well able to attempt part (a) with the result that many successful outcomes were seen. Part (b)(i) was only done correctly by the most able students, with $\int_0^t 4t$ $(1 - t^2)$ dt rarely seen otherwise. Many candidates either used 0 and 1 as their limits or simply ignored limits altogether. It was not sufficient to simply consider the indefinite integral $\int 4t$ $(1 - t^2)$ dt without consideration of the boundary conditions.

In part (b)(ii), many candidates realised that P ($\mu < T < m$) = F (m) – F (μ) but then went on to use the equation $2m^2 - m^4 = 0.5$ to show that $m = 0.541$ before attempting to evaluate $\int_{\frac{8}{15}}^{0.541} f(t)dt$, instead of simply evaluating 0.5 – F $\left(\frac{8}{15}\right)$.

Q14.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval [0 , 1] and the quadratic curve incorrectly drawn as convex on the interval [1 , 4]. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1. Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly with the correct answer of 0.85 often seen. Part (b)(iii) was not done well. Very few candidates tried to 'verify' the given result. Instead, most attempted to calculate q_x so that they could then show that its value fell within the given interval.

Q15.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval [0 , 1] and the quadratic curve incorrectly drawn as convex on the interval [1 , 4]. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1.

Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly. Several candidates wasted time in part (b)(iii) by considering the probability density function for $x > 1$ with the inevitable loss of most, if not all, of the available marks. Those candidates who did obtain a quadratic equation usually managed to solve it correctly and nearly always realised that one of their solutions was inadmissible. Part (b)(iv) was often done well, even by those candidates who failed to appreciate what was required in part (b)(iii).

Q16.

Most candidates knew how to use the distribution function to calculate probabilities and so parts (a) and (b)(i) caused few problems. However, there was some confusion in answering part (b)(ii). Some candidates were unsure which probability they were required to find whilst a few attempted to multiply probabilities for events that were not independent. It was pleasing to note that part (c) was generally well answered.

Q17.

In part (a) correct diagrams having axes marked with the pertinent values were the norm.

In part (b) $\frac{1}{54}t^3$ was frequently seen but only the most able candidates presented fully correct solutions. The vast majority of candidates simply did not know how to cope with finding F(t) when f(t) was defined in two intervals.

Part (c) was answered very well and by a variety of methods with the majority of candidates gaining full marks.

Q18.

Most candidates made a good start with this question and the results for the mean and standard deviation of the exponential distribution were well known. The majority of candidates used the given formula to calculate the probability in part (b) and to find the median in part (c). A few attempted both these parts by integrating the function they had written down in part (a)(i) and this wasted time.